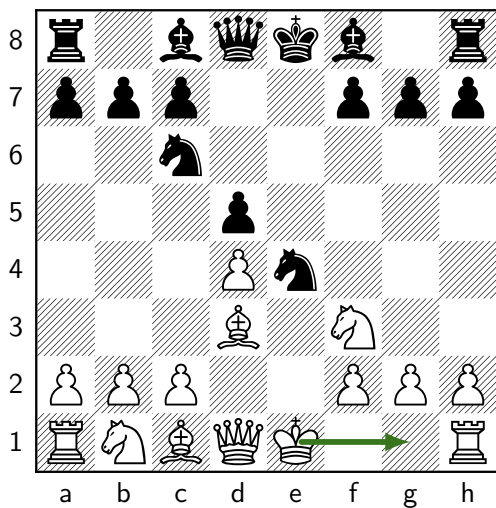


Melanie-ANALOGY — Petroff Defense

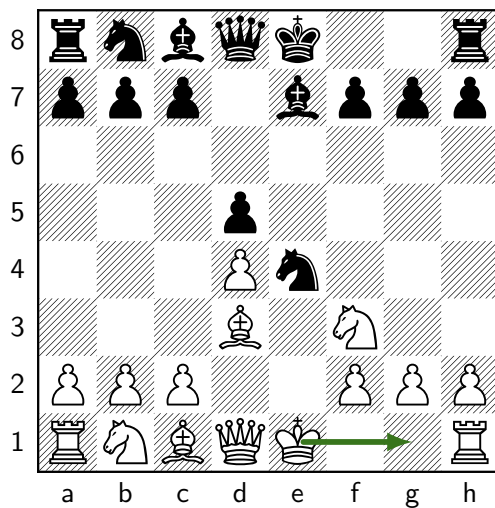
Melanie (Deep_Vision)

Abstract

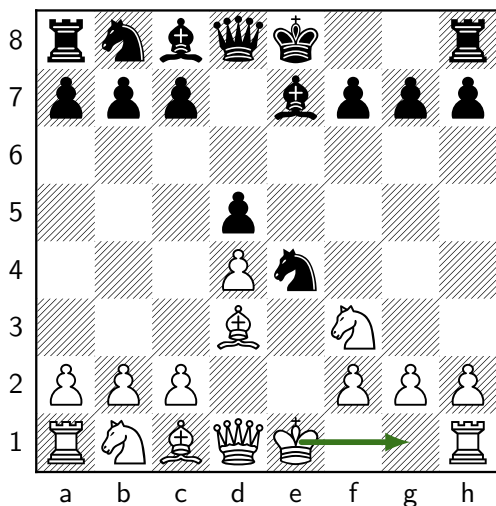
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



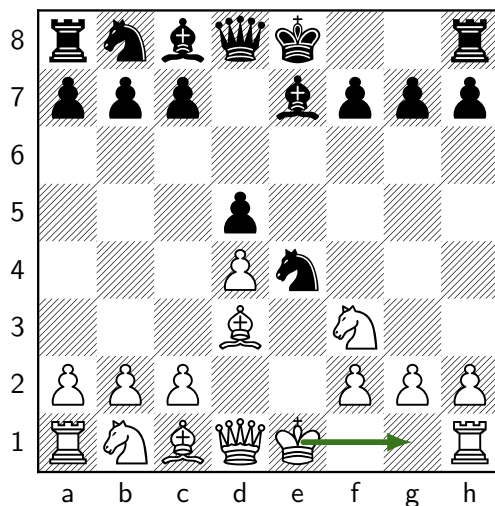
(a)



(b)

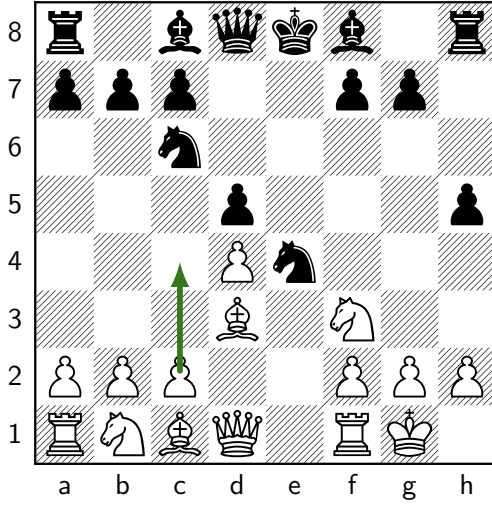


(c)

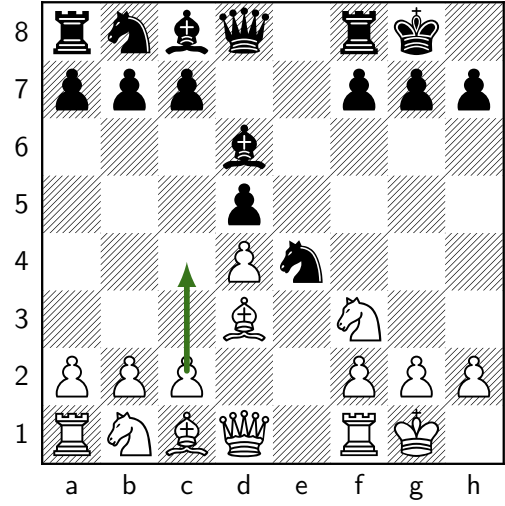


(d)

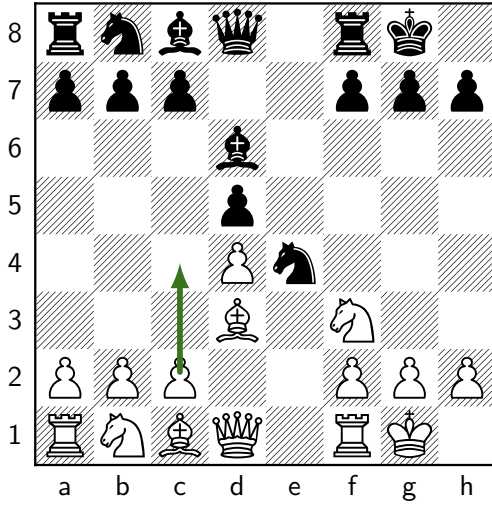
Figure 1: **Move 7, White: 0-0 (e1g1).** (a) Our position; Melanie plays 0-0 (e1g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2436` (ply 12) [structural distance $d=30$, QAP obj=51]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *nc6* as their *Ra1*, our *Pg2* as their *Bc1*; transferred move **e1g1 (endorses this move)**. (c) Retrieved from `ccrl_top_p0002#2437` (ply 12) [structural distance $d=30$, QAP obj=51]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *nc6* as their *Ra1*, our *Pg2* as their *Bc1*; transferred move **e1g1 (endorses this move)**. (d) Retrieved from `ccrl_top_p0003#629` (ply 12) [structural distance $d=30$, QAP obj=51]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *nc6* as their *Ra1*, our *Pg2* as their *Bc1*; transferred move **e1g1 (endorses this move)**. Synthesis: 18 retrieved analogs transferred to 0-0 (summed QAP objective 710, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



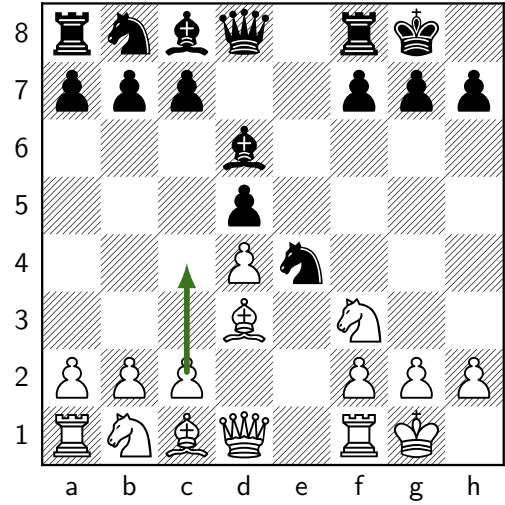
(a)



(b)

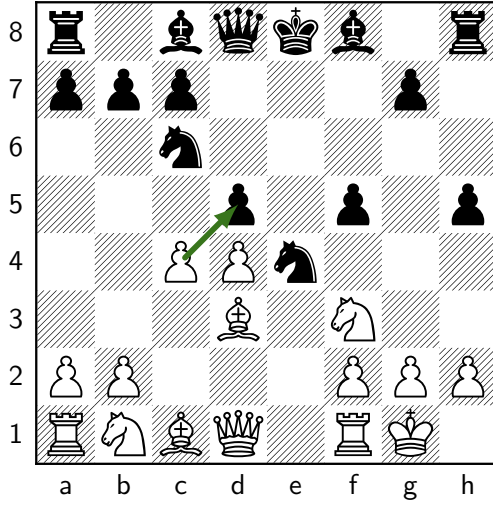


(c)

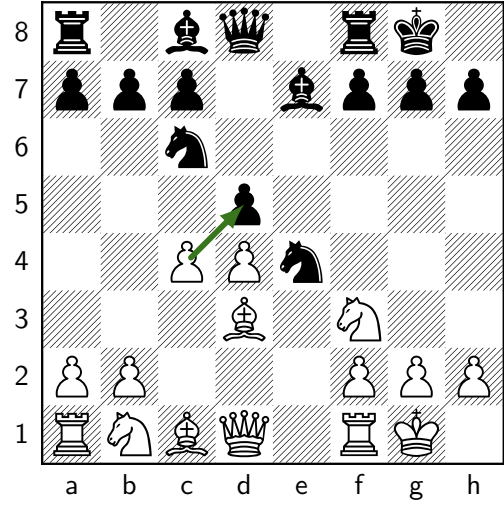


(d)

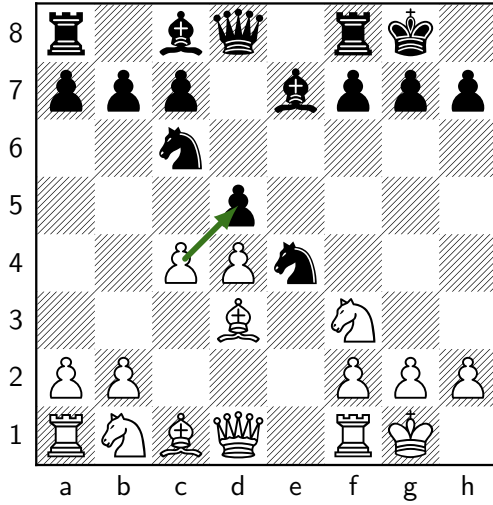
Figure 2: **Move 8, White: c4.** (a) Our position; Melanie plays c4 (c2c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3138` (ply 14) [structural distance $d=44$, QAP obj=53]: the master played c4; under μ our $Pc2$ plays the role of that game’s $Pc2$; cross-type roles: our $pg7$ as their $Ra1$, our $Pg2$ as their $Nb1$; transferred move c2c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3139` (ply 14) [structural distance $d=44$, QAP obj=53]: the master played c4; under μ our $Pc2$ plays the role of that game’s $Pc2$; cross-type roles: our $pg7$ as their $Ra1$, our $Pg2$ as their $Nb1$; transferred move c2c4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2991` (ply 14) [structural distance $d=44$, QAP obj=53]: the master played c4; under μ our $Pc2$ plays the role of that game’s $Pc2$; cross-type roles: our $pg7$ as their $Ra1$, our $Pg2$ as their $Nb1$; transferred move c2c4 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to c4 (summed QAP objective 426, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



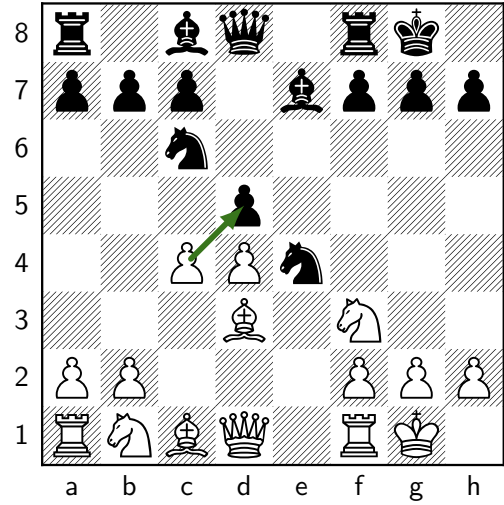
(a)



(b)

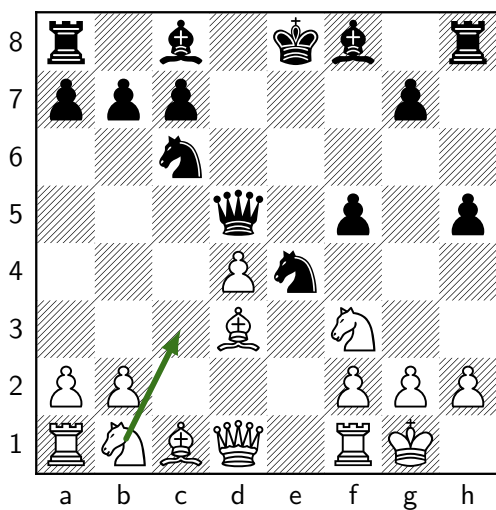


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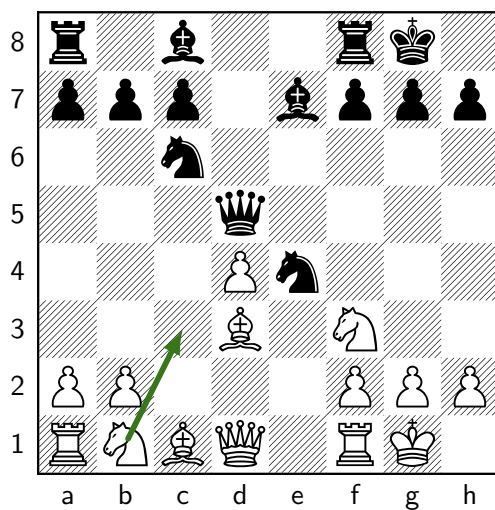


(d)

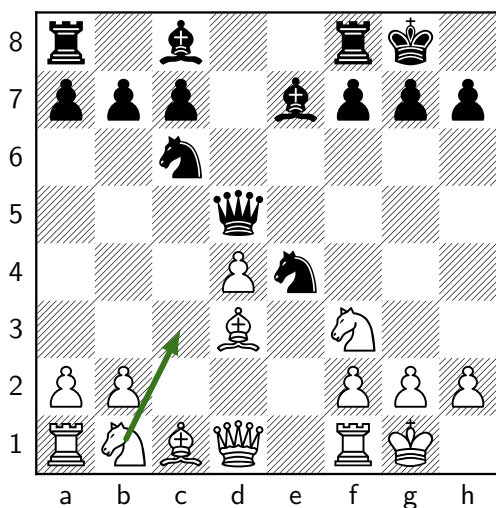
Figure 3: **Move 9, White: cxd5.** (a) Our position; Melanie plays cxd5 (c4d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#2556 (ply 16) [structural distance $d=42$, QAP obj=63]: the master played cxd5; under μ our $Pc4$ plays the role of that game’s $Pc4$; cross-type roles: our $Nb1$ as their $Ra1$, our $ra8$ as their $Nb1$; transferred move c4d5 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#2557 (ply 16) [structural distance $d=42$, QAP obj=63]: the master played cxd5; under μ our $Pc4$ plays the role of that game’s $Pc4$; cross-type roles: our $Nb1$ as their $Ra1$, our $ra8$ as their $Nb1$; transferred move c4d5 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#3890 (ply 16) [structural distance $d=42$, QAP obj=63]: the master played cxd5; under μ our $Pc4$ plays the role of that game’s $Pc4$; cross-type roles: our $Nb1$ as their $Ra1$, our $ra8$ as their $Nb1$; transferred move c4d5 (**endorses this move**). Synthesis: 30 retrieved analogs transferred to cxd5 (summed QAP objective 723, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



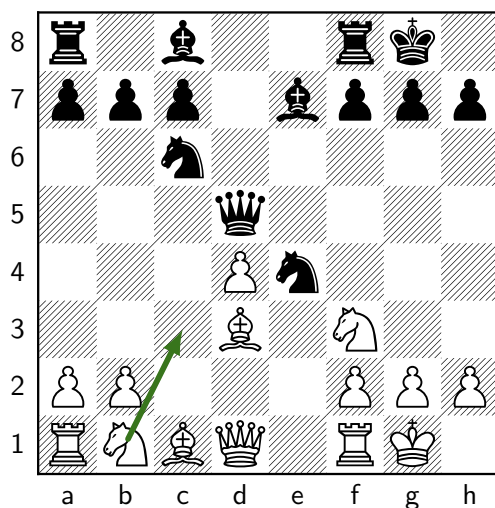
(a)



(b)

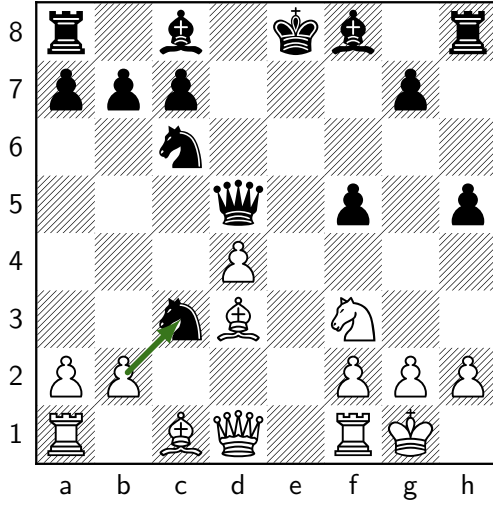


(c)

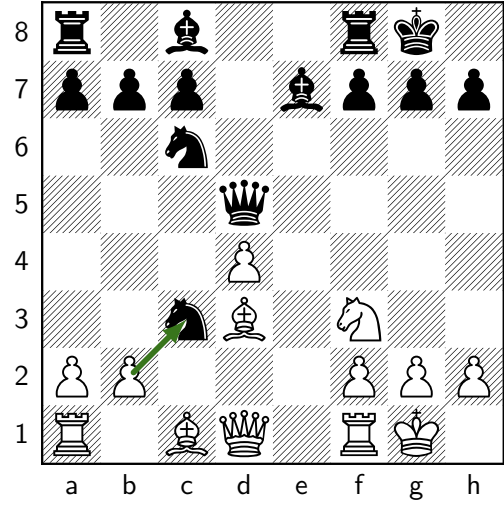


(d)

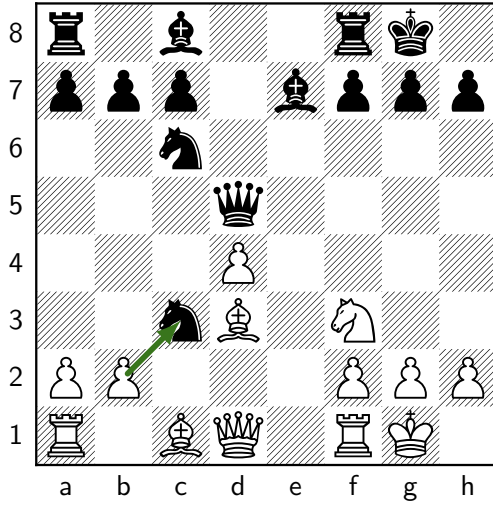
Figure 4: **Move 10, White: Nc3.** (a) Our position; Melanie plays Nc3 (b1c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2556` (ply 18) [structural distance $d=42$, QAP obj=121]: the master played Nc3; under μ our Nb1 plays the role of that game’s Nb1; cross-type roles: our ra8 as their Pb2, our pf5 as their be7; transferred move b1c3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2557` (ply 18) [structural distance $d=42$, QAP obj=121]: the master played Nc3; under μ our Nb1 plays the role of that game’s Nb1; cross-type roles: our ra8 as their Pb2, our pf5 as their be7; transferred move b1c3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3890` (ply 18) [structural distance $d=42$, QAP obj=121]: the master played Nc3; under μ our Nb1 plays the role of that game’s Nb1; cross-type roles: our ra8 as their Pb2, our pf5 as their be7; transferred move b1c3 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Nc3 (summed QAP objective 666, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



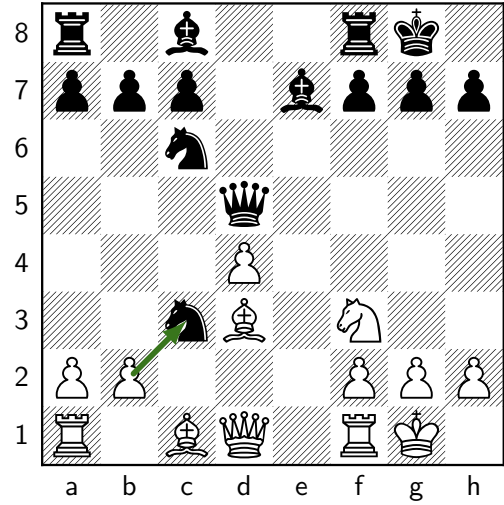
(a)



(b)

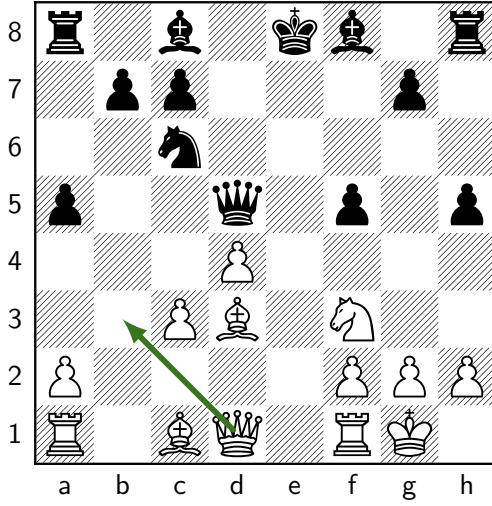


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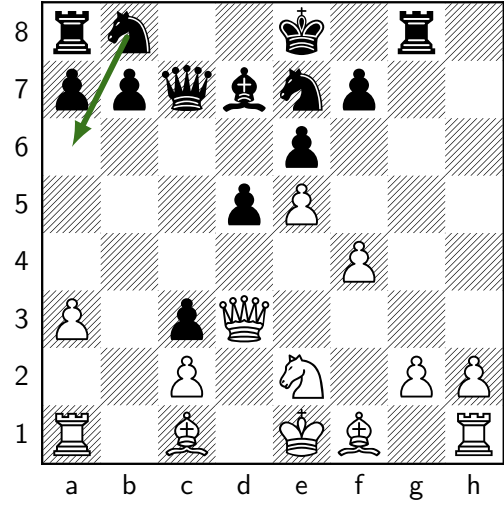


(d)

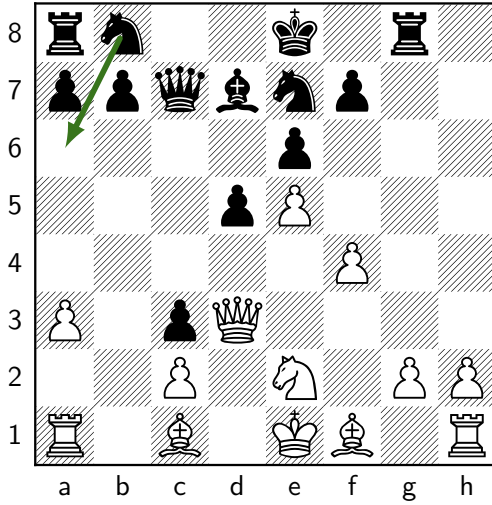
Figure 5: **Move 11, White: bxc3.** (a) Our position; Melanie plays **bxc3** (**b2c3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2556` (ply 20) [structural distance $d=35$, QAP obj=162]: the master played **bxc3**; under μ our *Pb2* plays the role of that game’s *Pb2*; cross-type roles: our *ra8* as their *Ph2*, our *bc8* as their *ph7*; transferred move **b2c3** (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2557` (ply 20) [structural distance $d=35$, QAP obj=162]: the master played **bxc3**; under μ our *Pb2* plays the role of that game’s *Pb2*; cross-type roles: our *ra8* as their *Ph2*, our *bc8* as their *ph7*; transferred move **b2c3** (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#3890` (ply 20) [structural distance $d=35$, QAP obj=162]: the master played **bxc3**; under μ our *Pb2* plays the role of that game’s *Pb2*; cross-type roles: our *ra8* as their *Ph2*, our *bc8* as their *ph7*; transferred move **b2c3** (**endorses this move**). Synthesis: 25 retrieved analogs transferred to **bxc3** (summed QAP objective 2201, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



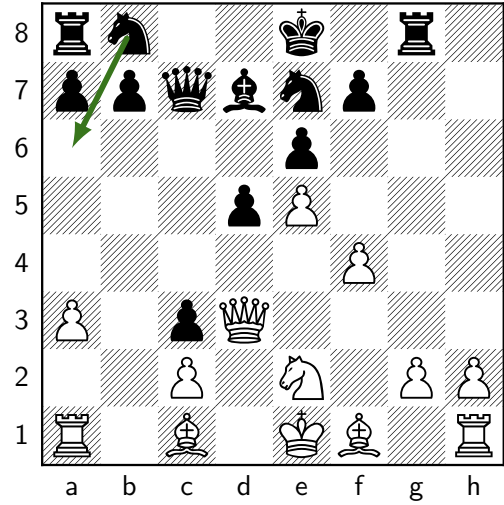
(a)



(b)

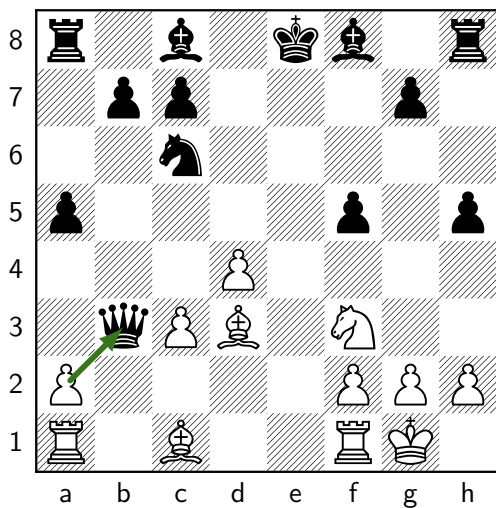


(c)

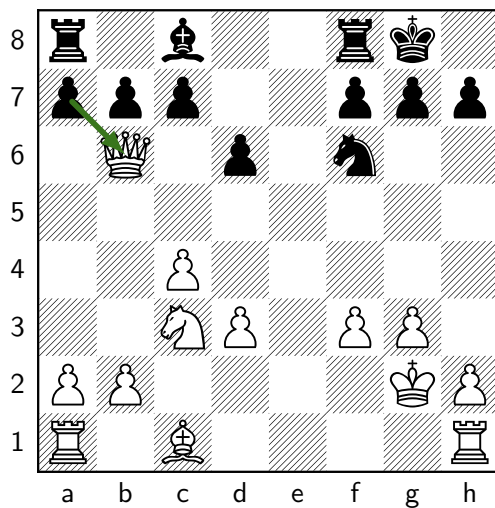


(d)

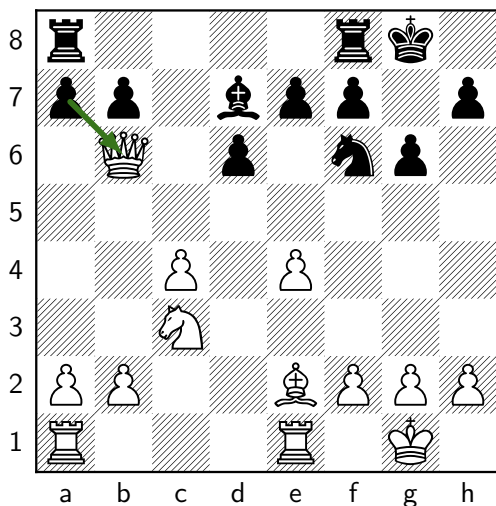
Figure 6: **Move 12, White: Qb3**. (a) Our position; Melanie plays Qb3 (d1b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2462` (ply 23) [structural distance $d=66$, QAP obj=59]: the master played Na6; under μ our Qd1 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our Bd3 as their Qc7, our pg7 as their Bd7; transferred move d1b3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2463` (ply 23) [structural distance $d=66$, QAP obj=59]: the master played Na6; under μ our Qd1 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our Bd3 as their Qc7, our pg7 as their Bd7; transferred move d1b3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2852` (ply 23) [structural distance $d=66$, QAP obj=59]: the master played Na6; under μ our Qd1 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our Bd3 as their Qc7, our pg7 as their Bd7; transferred move d1b3 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Qb3 (summed QAP objective 534, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



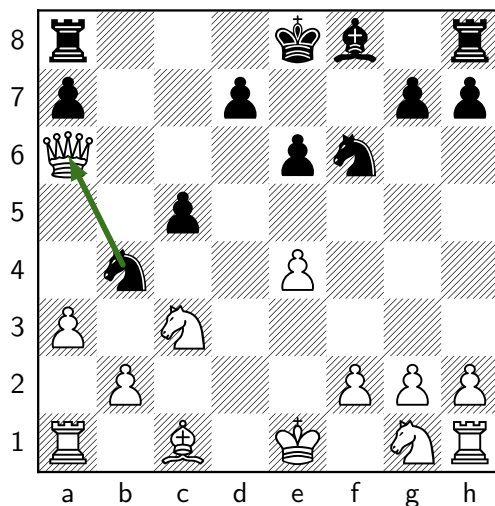
(a)



(b)

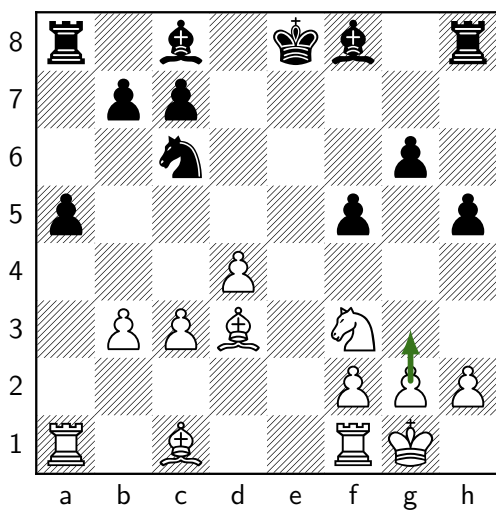


(c)

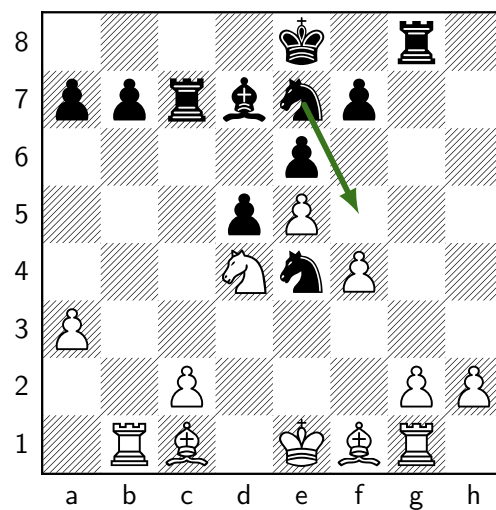


(d)

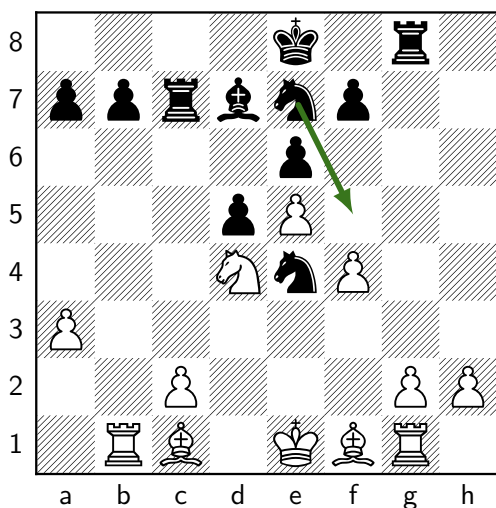
Figure 7: **Move 13, White: axb3.** (a) Our position; Melanie plays axb3 (a2b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#588 (ply 31) [structural distance $d=77$, QAP obj=92]: the master played axb6; under μ our Pa2 plays the role of that game’s Pa2; cross-type roles: our bf8 as their nc3, our rh8 as their ph2; transferred move a2b3 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#1252 (ply 25) [structural distance $d=81$, QAP obj=66]: the master played axb6; under μ our Pa2 plays the role of that game’s Pa2; cross-type roles: our nc6 as their pc4, our bc8 as their pe4; transferred move a2b3 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#2150 (ply 23) [structural distance $d=73$, QAP obj=63]: the master played Nxa6; under μ our Pa2 plays the role of that game’s Nb5 (a **cross-type** correspondence); cross-type roles: our Bd3 as their Pd7, our nc6 as their pe4; transferred move a2b3 (**endorses this move**). Synthesis: 27 retrieved analogs transferred to axb3 (summed QAP objective 1016, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



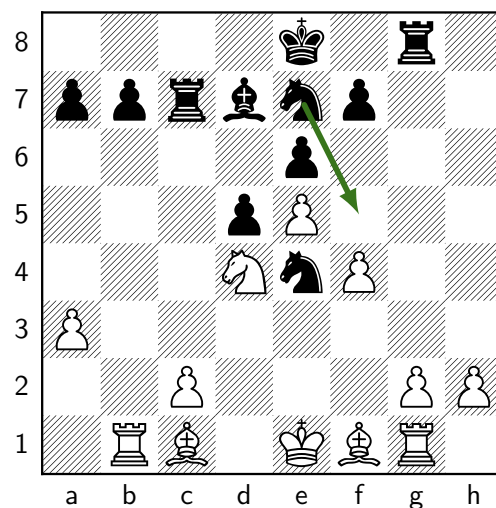
(a)



(b)

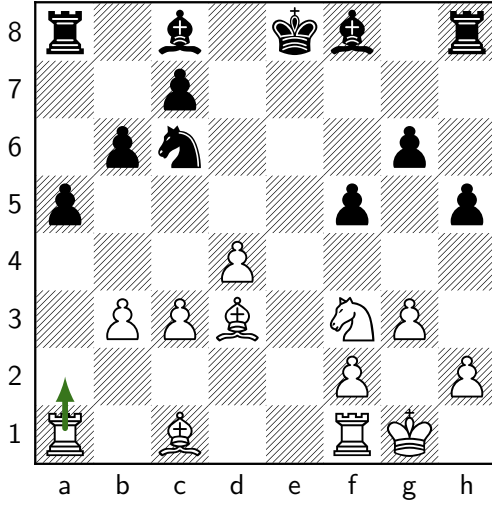


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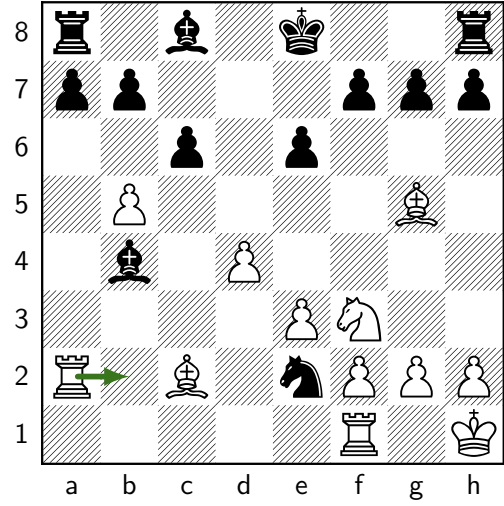


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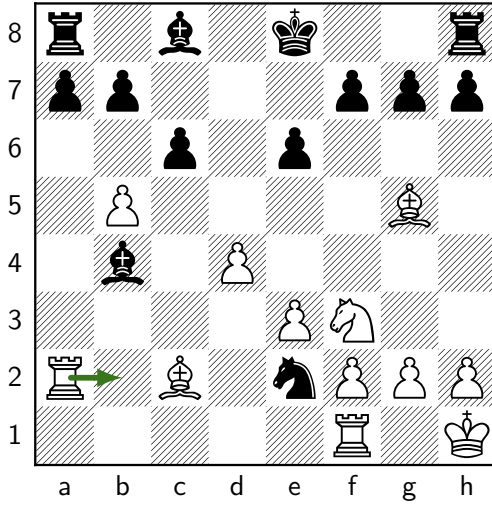
Figure 8: **Move 14, White: g3**. (a) Our position; Melanie plays **g3** (g2g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1245` (ply 33) [structural distance $d=54$, QAP obj=69]: the master played **Nf5**; under μ our *Pg2* plays the role of that game’s *Ne2* (a **cross-type** correspondence); cross-type roles: our *Nf3* as their *Bd7*, our *bf8* as their *Pf7*; transferred move **g2g3** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2852` (ply 33) [structural distance $d=54$, QAP obj=69]: the master played **Nf5**; under μ our *Pg2* plays the role of that game’s *Ne2* (a **cross-type** correspondence); cross-type roles: our *Nf3* as their *Bd7*, our *bf8* as their *Pf7*; transferred move **g2g3** (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#927` (ply 33) [structural distance $d=54$, QAP obj=69]: the master played **Nf5**; under μ our *Pg2* plays the role of that game’s *Ne2* (a **cross-type** correspondence); cross-type roles: our *Nf3* as their *Bd7*, our *bf8* as their *Pf7*; transferred move **g2g3** (**endorses this move**). Synthesis: 7 retrieved analogs transferred to **g3** (summed QAP objective 346, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



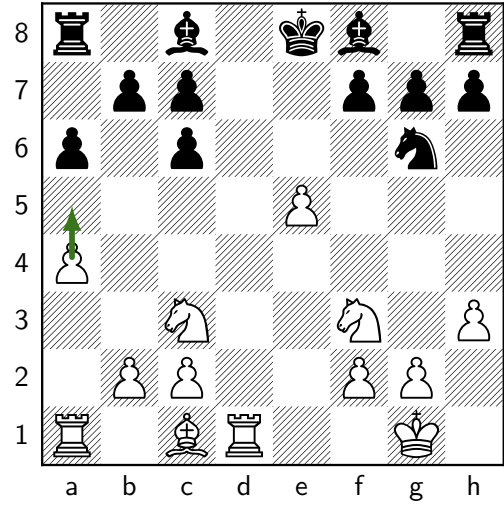
(a)



(b)

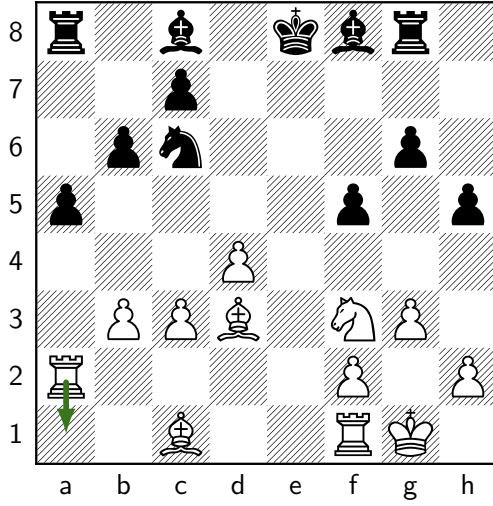


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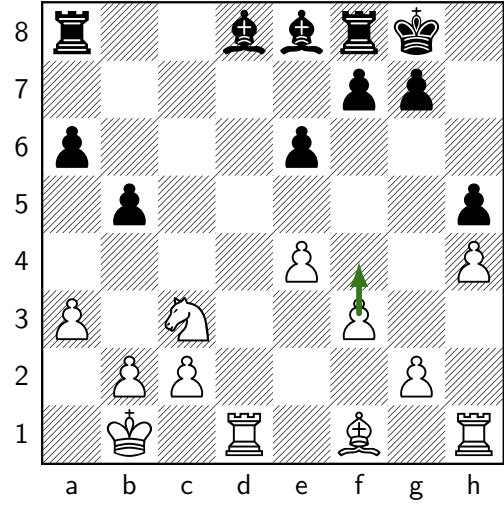


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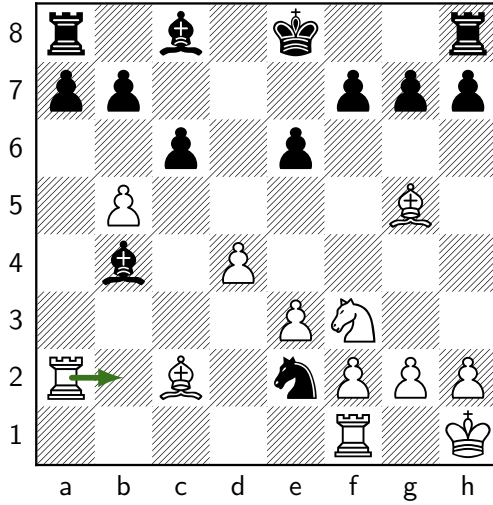
Figure 9: **Move 15, White: Ra2.** (a) Our position; Melanie plays Ra2 (a1a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1406` (ply 38) [structural distance $d=55$, QAP obj=43]: the master played Rb2; under μ our Ra1 plays the role of that game’s Ra2; cross-type roles: our Nf3 as their Rf1, our Rf1 as their Ph2; transferred move a1a2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1407` (ply 38) [structural distance $d=55$, QAP obj=43]: the master played Rb2; under μ our Ra1 plays the role of that game’s Ra2; cross-type roles: our Nf3 as their Rf1, our Rf1 as their Ph2; transferred move a1a2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#74` (ply 24) [structural distance $d=60$, QAP obj=29]: the master played a5; under μ our Ra1 plays the role of that game’s Pa4 (a **cross-type** correspondence); cross-type roles: our pg6 as their Ra1, our Pg3 as their Bc1; transferred move a1a2 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Ra2 (summed QAP objective 242, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



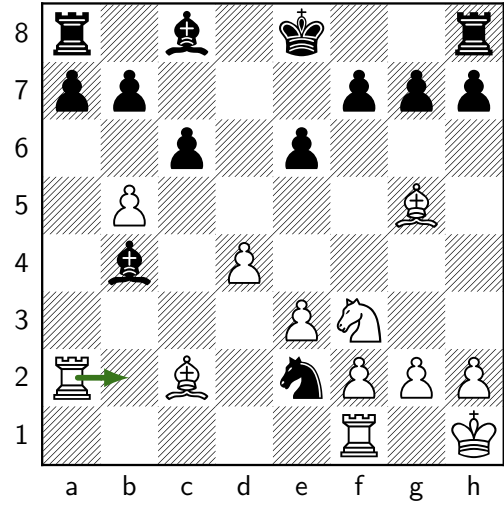
(a)



(b)

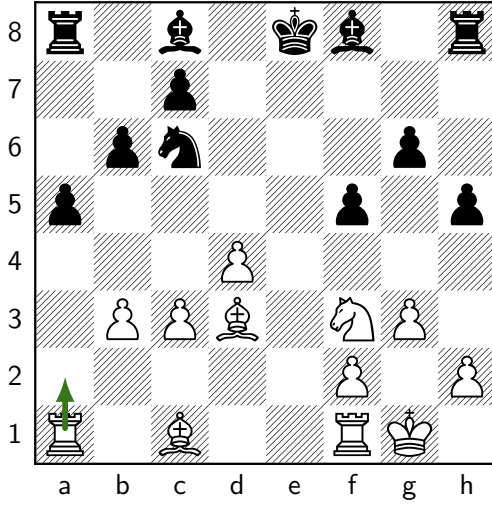


(c)

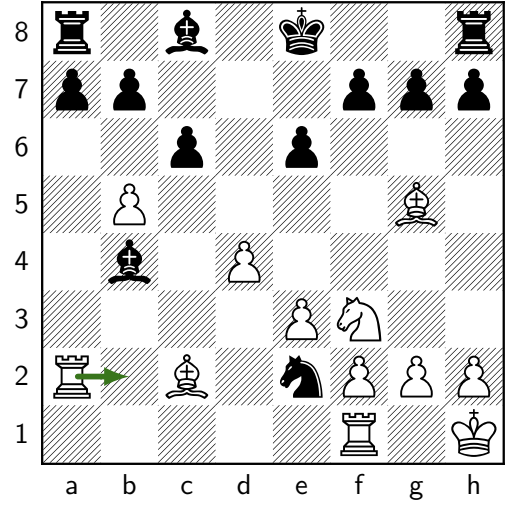


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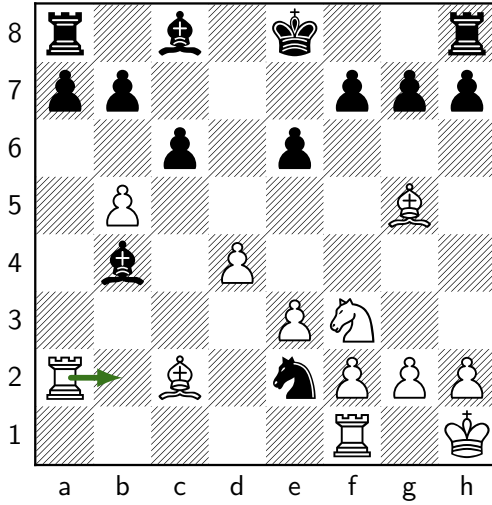
Figure 10: **Move 16, White: Ra1.** (a) Our position; Melanie plays Ra1 (a2a1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#930` (ply 32) [structural distance $d=66$, QAP obj=48]: the master played f4; under μ our *Ra2* plays the role of that game’s *Pf3* (a **cross-type** correspondence); cross-type roles: our *Bd3* as their *Rd1*, our *nc6* as their *Bf1*; transferred move a2a1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1406` (ply 38) [structural distance $d=57$, QAP obj=38]: the master played Rb2; under μ our *Ra2* plays the role of that game’s *Ra2*; cross-type roles: our *Nf3* as their *Rf1*, our *pc7* as their *Bc2*; transferred move a2a1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1407` (ply 38) [structural distance $d=57$, QAP obj=38]: the master played Rb2; under μ our *Ra2* plays the role of that game’s *Ra2*; cross-type roles: our *Nf3* as their *Rf1*, our *pc7* as their *Bc2*; transferred move a2a1 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Ra1 (summed QAP objective 235, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



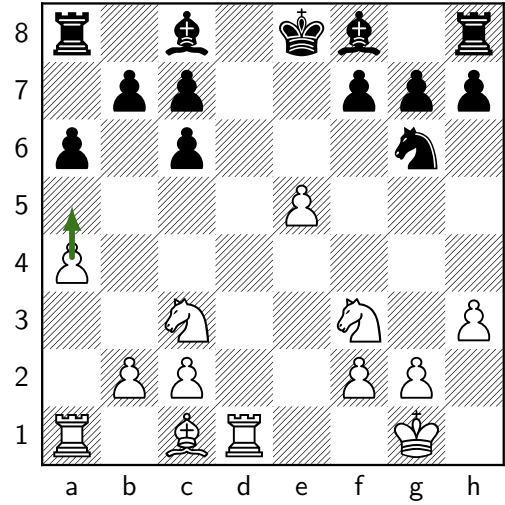
(a)



(b)

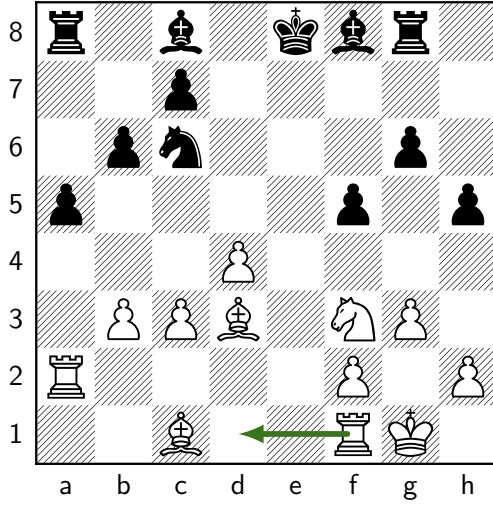


(c)

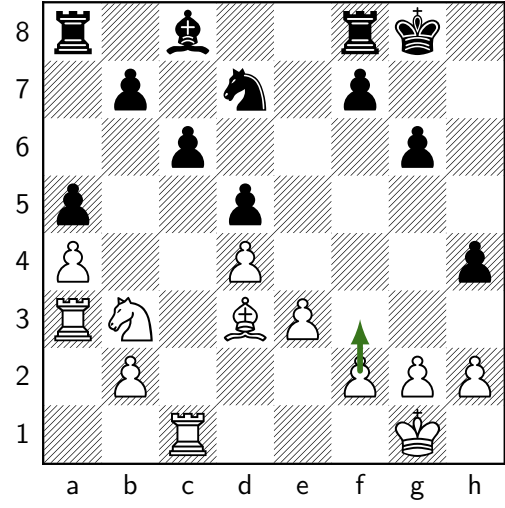


(d)

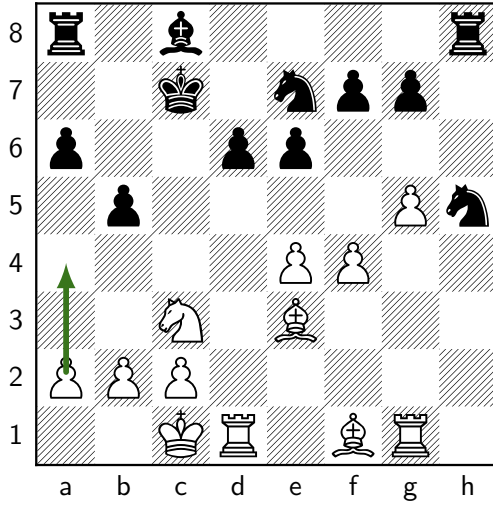
Figure 11: **Move 17, White: Ra2.** (a) Our position; Melanie plays Ra2 (a1a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1406` (ply 38) [structural distance $d=55$, QAP obj=43]: the master played Rb2; under μ our Ra1 plays the role of that game’s Ra2; cross-type roles: our Nf3 as their Rf1, our Rf1 as their Ph2; transferred move a1a2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1407` (ply 38) [structural distance $d=55$, QAP obj=43]: the master played Rb2; under μ our Ra1 plays the role of that game’s Ra2; cross-type roles: our Nf3 as their Rf1, our Rf1 as their Ph2; transferred move a1a2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#74` (ply 24) [structural distance $d=60$, QAP obj=29]: the master played a5; under μ our Ra1 plays the role of that game’s Pa4 (a **cross-type** correspondence); cross-type roles: our pg6 as their Ra1, our Pg3 as their Bc1; transferred move a1a2 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Ra2 (summed QAP objective 242, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



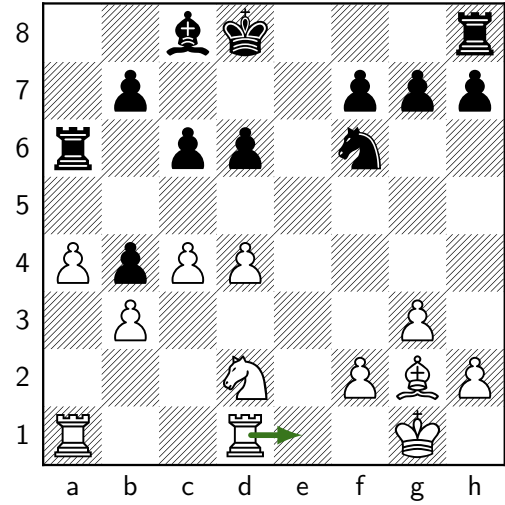
(a)



(b)

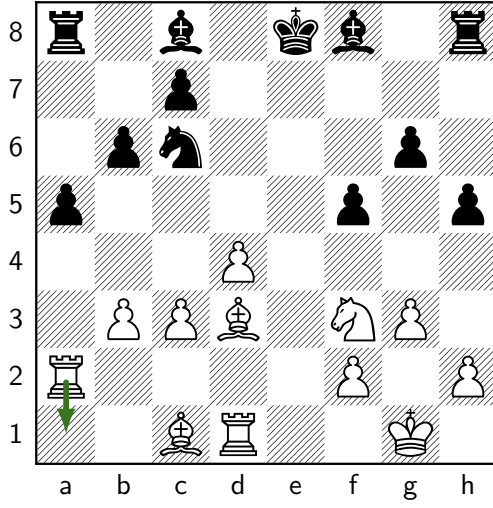


(c)

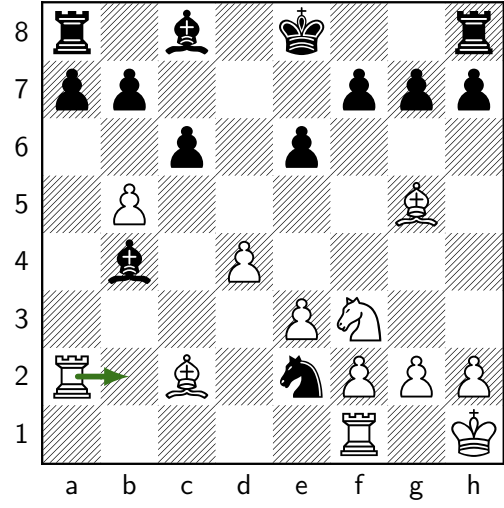


(d)

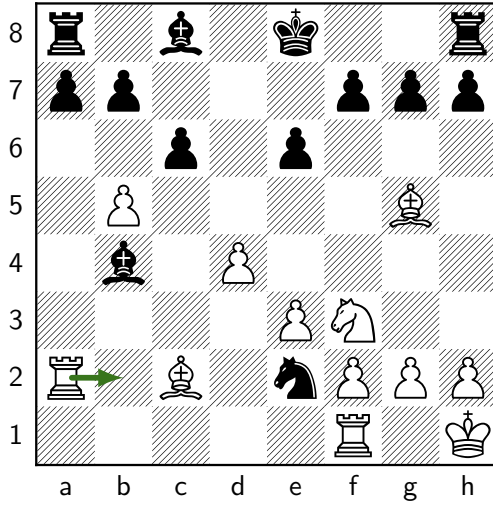
Figure 12: **Move 18, White: Rd1.** (a) Our position; Melanie plays Rd1 (f1d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3939` (ply 36) [structural distance $d=61$, QAP obj=55]: the master played f3; under μ our *Rf1* plays the role of that game’s *Pf2* (a **cross-type** correspondence); cross-type roles: our *Pg3* as their *Ra3*, our *Bd3* as their *Nb3*; transferred move f1d1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2654` (ply 42) [structural distance $d=49$, QAP obj=55]: the master played a4; under μ our *Rf1* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *Pb3* as their *Rg1*, our *Bc1* as their *Pb2*; transferred move f1d1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2661` (ply 32) [structural distance $d=60$, QAP obj=43]: the master played Re1; under μ our *Rf1* plays the role of that game’s *Rd1*; cross-type roles: our *pg6* as their *Nd2*, our *nc6* as their *Bg2*; transferred move f1d1 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Rd1 (summed QAP objective 240, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



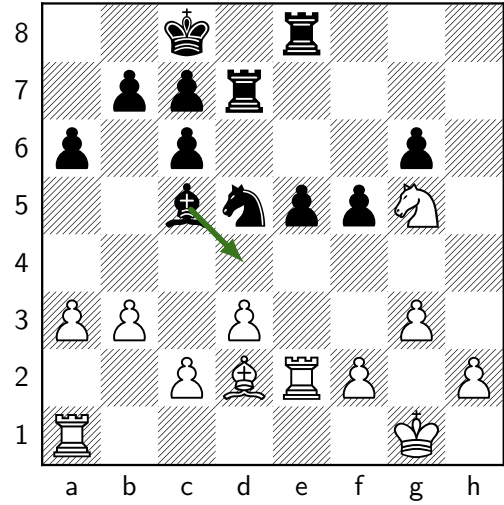
(a)



(b)

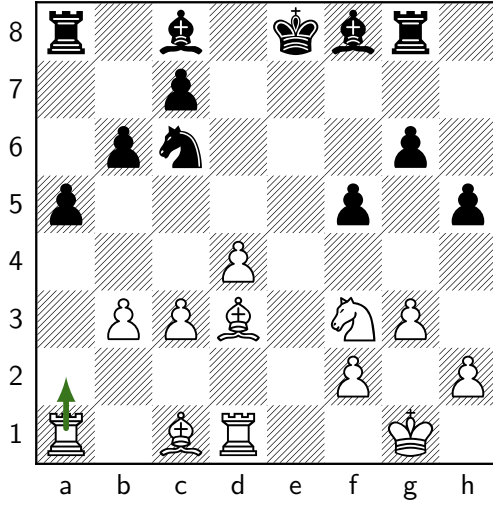


(c)

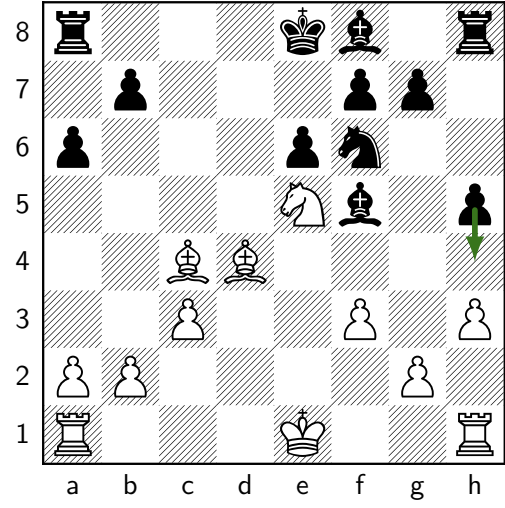


(d)

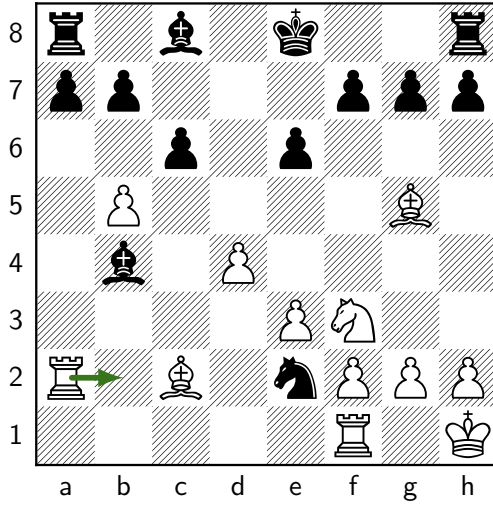
Figure 13: **Move 19, White: Ra1.** (a) Our position; Melanie plays Ra1 (a2a1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1406` (ply 38) [structural distance $d=48$, QAP obj=39]: the master played Rb2; under μ our Ra2 plays the role of that game's Ra2; cross-type roles: our Nf3 as their Rf1, our pc7 as their Bc2; transferred move a2a1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1407` (ply 38) [structural distance $d=48$, QAP obj=39]: the master played Rb2; under μ our Ra2 plays the role of that game's Ra2; cross-type roles: our Nf3 as their Rf1, our pc7 as their Bc2; transferred move a2a1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1154` (ply 43) [structural distance $d=62$, QAP obj=69]: the master played Bd4; under μ our Ra2 plays the role of that game's Bc4 (a **cross-type** correspondence); cross-type roles: our Nf3 as their Re8, our Bd3 as their Rd7; transferred move a2a4. Synthesis: 4 retrieved analogs transferred to Ra1 (summed QAP objective 154, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



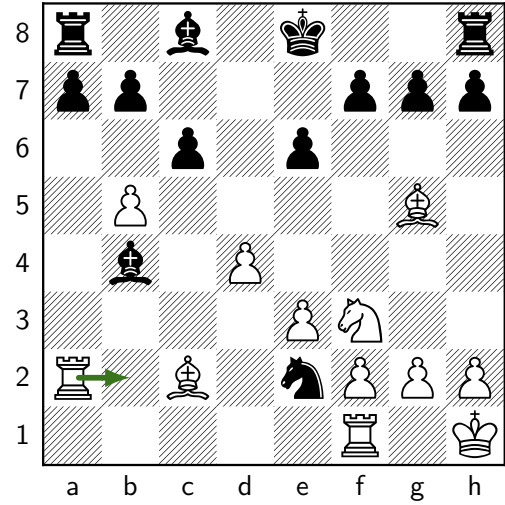
(a)



(b)

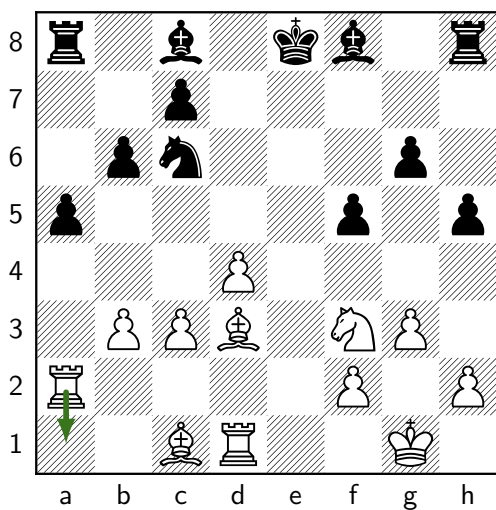


(c)

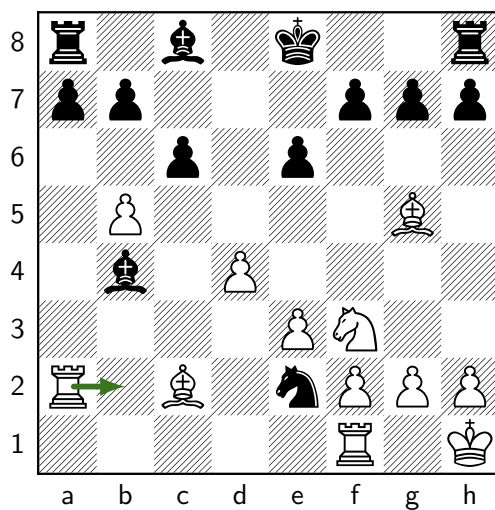


(d)

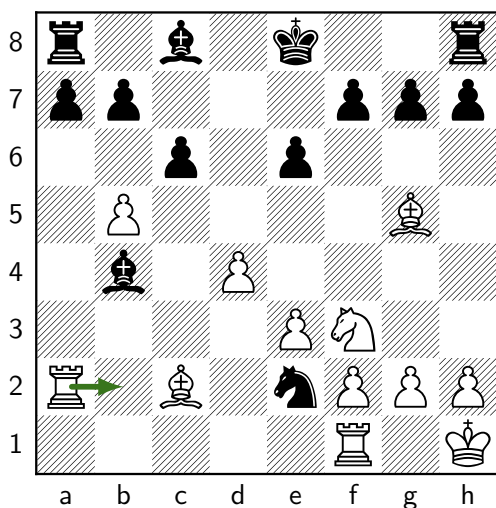
Figure 14: **Move 20, White: Ra2.** (a) Our position; Melanie plays Ra2 (a1a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#910` (ply 33) [structural distance $d=67$, QAP obj=70]: the master played **h4**; under μ our *Ra1* plays the role of that game’s *Ph4* (a **cross-type** correspondence); cross-type roles: our *Nf3* as their *Ra8*, our *Pg3* as their *Bf8*; transferred move **a1a2 (endorses this move)**. (c) Retrieved from `ccr1_top_p0002#1406` (ply 38) [structural distance $d=53$, QAP obj=45]: the master played **Rb2**; under μ our *Ra1* plays the role of that game’s *Ra2*; cross-type roles: our *Nf3* as their *Rf1*, our *Rd1* as their *Ph2*; transferred move **a1a2 (endorses this move)**. (d) Retrieved from `ccr1_top_p0002#1407` (ply 38) [structural distance $d=53$, QAP obj=45]: the master played **Rb2**; under μ our *Ra1* plays the role of that game’s *Ra2*; cross-type roles: our *Nf3* as their *Rf1*, our *Rd1* as their *Ph2*; transferred move **a1a2 (endorses this move)**. Synthesis: 6 retrieved analogs transferred to Ra2 (summed QAP objective 268, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



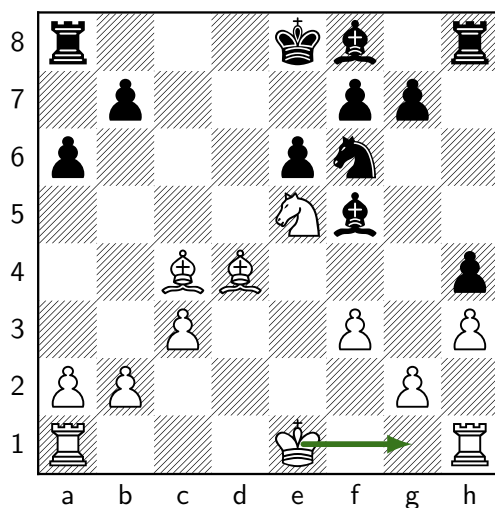
(a)



(b)



(c)



(d)

Figure 15: **Move 21, White: Ra1.** (a) Our position; Melanie plays Ra1 (a2a1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1406` (ply 38) [structural distance $d=48$, QAP obj=39]: the master played Rb2; under μ our Ra2 plays the role of that game’s Ra2; cross-type roles: our Nf3 as their Rf1, our pc7 as their Bc2; transferred move a2a1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1407` (ply 38) [structural distance $d=48$, QAP obj=39]: the master played Rb2; under μ our Ra2 plays the role of that game’s Ra2; cross-type roles: our Nf3 as their Rf1, our pc7 as their Bc2; transferred move a2a1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#910` (ply 34) [structural distance $d=67$, QAP obj=70]: the master played 0-0; under μ our Kg1 plays the role of that game’s Ke1; cross-type roles: our Bc1 as their Pf3, our nc6 as their Bc4; transferred move g1f1. Synthesis: 4 retrieved analogs transferred to Ra1 (summed QAP objective 154, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.